

Heart Failure Risk Prediction in Smurfs: A Linear Modeling Approach

zeamari yassine — Rajet Jebri — group: 53

1 Introduction

Heart failure is a major health concern in Smurf society influenced by biological, characteristics, lifestyle factors and cardiac morphology. The goal of this project is to build a predictive model capable of estimating heart failure risk using both tabular data and heart scan images.

The work is divided into four parts. We first analyze the tabular data using linear models and feature-selection techniques to identify the most important predictors. We then evaluate several non-linear machine learning models and select the one that offers the best performance. In the third part, we integrate information from heart images by training a CNN to extract meaningful features, which are then combined with the tabular variables. Finally, we study the broader factors influencing heart failure risk, using visualizations and clustering to identify groups of Smurfs most at risk.

2 Part 1

2.1 Data Preprocessing

We first performed a complete cleaning of the dataset. No duplicated rows were found among the 1000 observations so no removal was required. A missing-value analysis confirmed that all variables (numerical and categorical) contain 0 missing entries, meaning no imputation was necessary.

Outlier detection was conducted using the Z-score method (threshold $|Z| > 3$) and IQR method ($[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$) on all numerical features. Z-score method Detected 43 outliers total, with blood pressure (20), potassium (4), and hemoglobin (4) showing the most extreme values.

while IQR Method Detected 96 outliers total, with blood pressure (34), vitamin D (18), and potassium (9) most affected. IQR is more sensitive as it doesn't assume normality.

Based on the above diagnostic, the following pipeline was applied:

- **Outlier Treatment:** Winsorization using IQR method to cap extreme values
- **Categorical Encoding:** One-hot encoding for profession and consumption levels
- **Scaling:** StandardScaler for numerical features

This preprocessing generates a clean and model-ready dataset compatible with linear regression techniques.

Unexpected Impact of Outlier Treatment:

Table 1: Impact of Outlier Treatment

Metric	Raw	Treated	Impact	
CV (Ridge, k=5)				
RMSE	0.0558	0.0575	+0.0017	↑ 3%
R ²	0.5170	0.4886	-0.0284	↓ 5.5%
Test Set (20%)				
RMSE	0.0507	0.0512	+0.0005	↑ 1%
R ²	0.4854	0.4757	-0.0097	↓ 2%
Training Set				
RMSE	0.0539	0.0558	+0.0019	↑ 3.5%
R ²	0.5548	0.5233	-0.0315	↓ 5.7%
AIC	-2953	-2887	+66	Worse
BIC	-2831	-2769	+62	Worse

Outlier treatment degraded model performance. RMSE increased 3% in CV and 1% on test data, while R² decreased 5.5% and 2% respectively.

Explanation: Detected "outliers" likely contain valuable predictive information:

- Extreme clinical values may genuinely indicate risk
- Winsorization compressed feature distributions
- Linear models benefit from full value ranges
- Medical data contains legitimate biological variability

In our dataset outliers were informative rather than erroneous. Their removal reduced the predictive power of linear models.

2.2 Feature Selection

With cleaned and preprocessed data we proceed to evaluate multiple feature selection strategies (filter, wrapper, and embedded methods). The goal is to identify the most relevant predictors for heart failure risk prior to linear modeling.

Table 2: Comparison of Feature Selection Methods (sorted by RMSE)

Method	k retained	RMSE (CV=5)	R ² (CV=5)
Embedded - LASSO	24	0.0559	0.5151
Wrapper - Backward	29	0.0559	0.5141
Wrapper - Forward	29	0.0560	0.5135
correlation	30	0.0560	0.5135
mRMR (custom)	15	0.0565	0.5026

five feature selection methods were evaluated: Correlation filtering, Sequential Forward Selection, Sequential Backward Elimination, LASSO (embedded), and a custom mRMR variant. The comparison, summarized in Table 2, shows that the **LASSO** method achieved the best overall performance, with the lowest cross-validated RMSE (0.0559) and the highest R² (0.515).

While correlation filtering, and Sequential Forward/Backward Selection produced similar predictive performance (RMSE around 0.0560), these methods retained a substantially larger number of features (29–30).

The custom mRMR approach, although theoretically attractive for balancing redundancy and relevance, yielded slightly worse results (RMSE = 0.0565)

Conclusion:

LASSO provides the best compromise between predictive accuracy and model interpretability by retaining only 24 variables while maintaining the strongest validation performance.

2.3 Model Selection

Evaluated Linear Regression, Ridge, LASSO, and ElasticNet using 5-fold cross-validation, hold-out validation (80/20 split), AIC/BIC scores and RMSE.

Model	RMSE (train)	R ² (train)	AIC (train)	BIC (train)
Ridge ($\alpha = 1.0$)	0.053908	0.554838	-2953.131	-2830.611
Linear Regression	0.053908	0.554838	-2953.061	-2830.367
LASSO (CV)	0.053958	0.554019	-2951.221	-2828.527
ElasticNet (CV)	0.053961	0.553965	-2951.101	-2828.407

Table 3: Comparison of Linear Models

Table 4: Hold-out Validation (80/20 Split)

Model	RMSE (hold-out 20%)	R ² (hold-out 20%)
Ridge ($\alpha = 1.0$)	0.050688	0.485377
Linear Regression	0.050696	0.485213
LASSO (CV)	0.050730	0.484523
ElasticNet (CV)	0.050737	0.484379

Table 5: K-Fold Cross-Validation Results (k=5)

Model	RMSE (CV=5)	R ² (CV=5)
ElasticNet (CV)	0.055749	0.517129
LASSO (CV)	0.055750	0.517114
Ridge ($\alpha = 1.0$)	0.055765	0.516950
Linear Regression	0.055769	0.516878

The results in Tables 3- 4-5 show that all linear models — Linear Regression, Ridge, LASSO, and ElasticNet — give almost the same performance. This means the data has mainly linear relationships. Ridge and Linear Regression have the lowest AIC and BIC values, showing a good balance between accuracy and model size. In the 80/20 validation, Ridge gives the best RMSE (0.0507) and R^2 (0.485), so it generalizes well. In 5-fold cross-validation, all models perform almost the same, with LASSO and ElasticNet slightly better. In this part, we use RMSE (Root Mean Squared Error) as the main evaluation criterion, LASSO was chosen as the final model based on its good RMSE and feature selection ability.

The LASSO regression selected 24 variables that best explain the heart failure risk. 6 Among them, blood pressure shows the strongest positive influence, meaning higher blood pressure increases the predicted risk. Other important positive contributors include weight, age, cholesterol, and potassium, all associated with higher cardiovascular stress. Conversely, features such as hemoglobin, height, and high sarsaparilla consumption have negative coefficients, suggesting inverse relationship with the risk score.

Table 6: LASSO Selected Feature Coefficients

Feature	Coefficient
blood pressure	0.039539
weight	0.011314
age	0.009843
cholesterol	0.007740
potassium	0.005948
sarsaparilla_Very low	0.003780
smurfin donuts_Very high	0.002911
profession_administration and governance	0.002832
sarsaparilla_Low	0.001954
profession_services	0.001407
smurfberry liquor_Very low	0.000912
calcium	0.000782
smurfberry liquor_Moderate	0.000619
smurfin donuts_Very low	0.000608
vitamin D	-0.000655
smurfberry liquor_Low	-0.001161
smurfberry liquor_Very high	-0.001761
profession_food production	-0.001817
smurfin donuts_Moderate	-0.001839

3 Part 2

3.1 Feature selection for nonlinear models

We compared four feature selection strategies on the tabular data:

- **Embedded – Random Forest importance:** kept only the 6 most important variables (blood pressure, hemoglobin, potassium, weight, age, cholesterol). This produced a compact model but test performance was lower (RF: RMSE $\approx$ 0.0467, $R^2 \approx$ 0.676).
- **Wrapper – RFE with Random Forest:** recursively removed features until 15 variables remained. Test performance improved slightly (RMSE $\approx$ 0.0459, $R^2 \approx$ 0.688), but at the cost of a larger feature set.
- **Embedded – XGBoost importance:** features were ranked by gain and we kept the smallest subset explaining 90% of the total gain (22 variables, mixing clinical and lifestyle features). This configuration gave the **best overall performance** (XGBoost: RMSE $\approx$ 0.0434, $R^2 \approx$ 0.720).
- **Filter – Mutual Information:** selected 10 variables explaining 95% of the mutual information with the target. However, the resulting Random Forest was clearly worse (RMSE $\approx$ 0.0511, $R^2 \approx$ 0.612).

Overall, the XGBoost-based embedded method offers the best trade-off between accuracy and model richness. It achieves the lowest RMSE and highest R^2 while keeping a reasonably sized set of 22 interpretable features (blood pressure, cholesterol, hemoglobin, potassium, weight, age, several consumption levels and professions).

For this reason, we use the **XGBoost-selected features as the common input** for all nonlinear models in Part 2 (Random Forest, XGBoost, SVR and MLP).

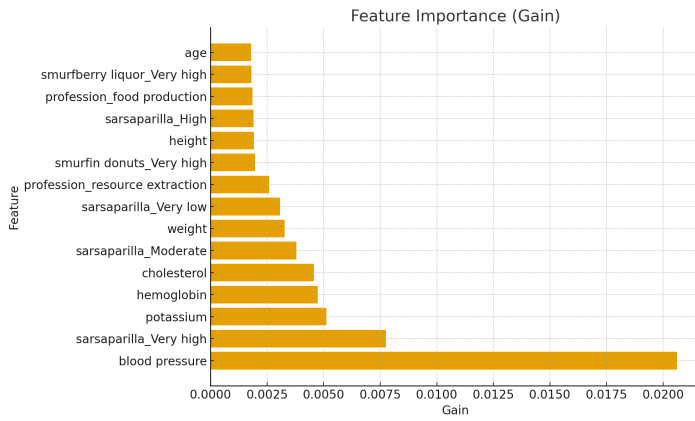


Figure 1: features are most important

Across all nonlinear feature selection methods a consistent set of variables emerges as the strongest predictors of heart failure risk:

blood pressure and sarsaparilla consumption (Very high),potassium,hemoglobin,cholesterol and weight.

Blood pressure stands out dramatically with a gain score 3× higher than any other feature making it the most influential factor in the nonlinear model.

These clinical variables were also important in Part 1 confirming their robustness.

However unlike linear models nonlinear methods also highlight lifestyle (sarsaparilla, smurfin donuts...) and profession categories as relevant predictors suggesting the presence of nonlinear interactions and threshold effects.

The preprocessing pipeline was adapted accordingly:feature selection redesigned using nonlinear criteria. The XGBoost-selected feature set produced the best overall performance (RMSE = 0.043, R^2 = 0.72)

3.2 Hyper-parameter tuning for nonlinear models

For all nonlinear models we did model selection by cross-validated hyper-parameter search.We used Randomized-SearchCV with 5-fold CV and the score RMSE.

The search was done only on the training set with tabular features selected by XGBoost, and the test set was used once at the end to estimate generalization.

Table 7: Best hyperparameters obtained via 5-fold RandomizedSearchCV

Model	Best Hyperparameters
Random Forest	$n_estimators = 600$ $max_depth = 15$ $min_samples_split = 5$ $min_samples_leaf = 1$ $max_features = 0.8$
XGBoost	$n_estimators = 500$ $learning_rate = 0.01$ $max_depth = 5$ $subsample = 0.8$ $colsample_bytree = 0.6$ $reg_lambda = 0.0$ $reg_alpha = 0.0$
SVR (RBF)	$C = 1$ $\gamma = 0.01$ $\epsilon = 0.01$
MLP	$hidden_layer_sizes = (64, 32)$ $\alpha = 0.01$ $learning_rate_init = 0.01$

3.3 Comparison of Nonlinear Models

Four nonlinear models were evaluated: Random Forest, XGBoost, SVR (RBF kernel), and a Multilayer Perceptron (MLP). Model selection was performed using both a validation split (80/20) and 5-fold cross-validation. Final performance was assessed on the provided test set

Validation Split (80/20)

Model	RMSE (val)	R^2 (val)
XGBoost	0.03923	0.6917
Random Forest	0.04225	0.6424
SVR (RBF)	0.04283	0.6326
MLP	0.06004	0.2779

Table 8: Validation split performance (80/20).

5-Fold Cross-Validation

Model	RMSE (CV)	R^2 (CV)
XGBoost	0.04174	0.7268
Random Forest	0.04364	0.7010
SVR (RBF)	0.04417	0.6946
MLP	0.05934	0.4375

Table 9: 5-fold cross-validation results.

Final Test Set Performance

Model	RMSE (test)	R^2 (test)
XGBoost	0.04347	0.7196
SVR (RBF)	0.04542	0.6940
Random Forest	0.04641	0.6805
MLP	0.06278	0.4154

Table 10: Final test set performance.

Across all evaluation criteria, XGBoost consistently achieves the best performance. It obtains the lowest RMSE and highest R^2 on the validation split, cross-validation, and final test set. Random Forest and SVR provide competitive but slightly inferior results, while the MLP suffers from instability and overfitting, leading to significantly lower predictive accuracy.

Therefore, XGBoost is selected as the **best nonlinear model**, offering the strongest generalization capability and the most reliable performance across all metrics.

4 Part 3

we integrate image-derived features into our prediction pipeline in order to build a multimodal model combining clinical and visual information.

4.1 Methodology

The objective is to transform each raw image into a compact numerical representation that can be fused with tabular clinical data.

4.1.1 Image Preprocessing

All images undergo a standardized preprocessing pipeline implemented in the function `default_transform`. The following operations are applied:

1. **Grayscale conversion:** All images are converted to a single intensity channel:

$$I = \text{convert_to_L}(\text{image}).$$

2. **Spatial normalization:** Images are resized to a fixed resolution of **48×48 pixels**.
3. **Intensity normalization:** Pixel intensities are scaled to the range $[0, 1]$ and standardized to $[-1, 1]$:

$$I_{\text{norm}} = \frac{I/255 - 0.5}{0.5}.$$

4. **Tensor formatting:** Images are reshaped to PyTorch format:

$$(1, 48, 48).$$

4.1.2 Dataset Structure

A custom dataset class (`CustomDataset`) loads images from disk, applies preprocessing, and optionally returns the associated risk label.

4.1.3 CNN Architecture (SmurfCNN)

The proposed CNN is lightweight and adapted to the small input size. It consists of three convolutional blocks with dropout for regularization:

Conv2d → ReLU → MaxPool2d → Dropout,
 Conv2d → ReLU → MaxPool2d → Dropout,
 Conv2d → ReLU → AdaptiveAvgPool2d(3×3).

The final convolutional output has dimension $64 \times 3 \times 3$, which is flattened and projected onto a compact embedding space using a fully connected layer:

$$\mathbf{e} = \text{ReLU}(Wz), \quad \mathbf{e} \in R^d.$$

A final linear layer predicts the risk score during training:

$$\hat{y} = w^\top \mathbf{e}.$$

During feature extraction, only $\mathbf{e}$ is retained.

4.1.4 Training Procedure

The CNN is trained using the following configuration:

- 75% of the images used for training, 25% for validation.
- Loss: Mean Squared Error (MSE).
- Optimizer: Adam ($\text{lr} = 5 \cdot 10^{-4}$).
- Batch size: 64.
- Epochs: 30.

For each epoch, the algorithm performs:

1. Forward propagation on the training set.
2. Backpropagation and parameter update.
3. Evaluation on the validation set.

Both Train RMSE and Validation RMSE are logged to monitor convergence and overfitting.

4.1.5 Feature Extraction

After training, the prediction head is ignored and the CNN is used solely as a feature extractor. Given a set of n images, the method `extract_features` returns a matrix:

$$Z = \begin{pmatrix} \mathbf{e}_1^\top \\ \vdots \\ \mathbf{e}_n^\top \end{pmatrix} \in R^{n \times d},$$

where d denotes the embedding dimension 32. These embeddings are standardized using `StandardScaler` and concatenated with tabular features.

4.2 Hyperparameter Sensitivity Analysis of the CNN Image Encoder

4.2.1 Impact of Embedding Dimension (n_{features})

The embedding produced by the CNN is a compact representation of each heart image. We tested four configurations: 16, 32, 64, and 128 features.

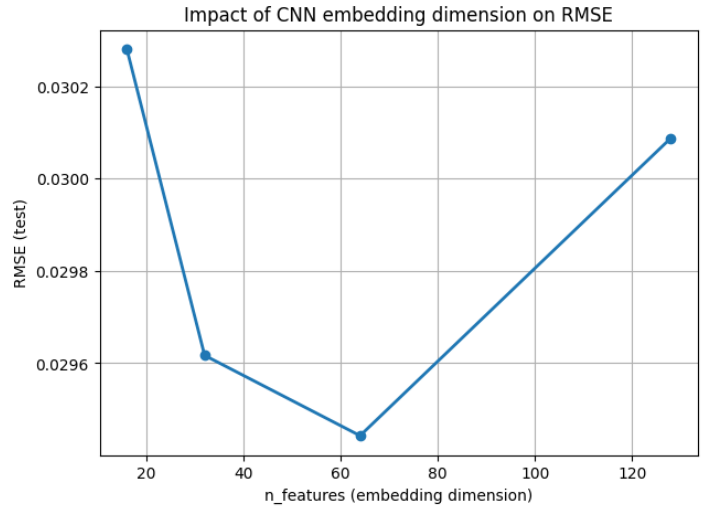


Figure 2: Impact of Embedding Dimension

Increasing `nfeatures` improves representation power up to 64 dimensions.

Beyond this point (e.g. 128) performance drops slightly indicating: Overfitting to the training images or Increased difficulty for XGBoost to exploit excessively large embeddings

4.2.2 Impact of Training Epochs

We also varied the training duration of the CNN:

5 Part 4

5.1 t-SNE Visualization Reveals Distinct Risk Clusters

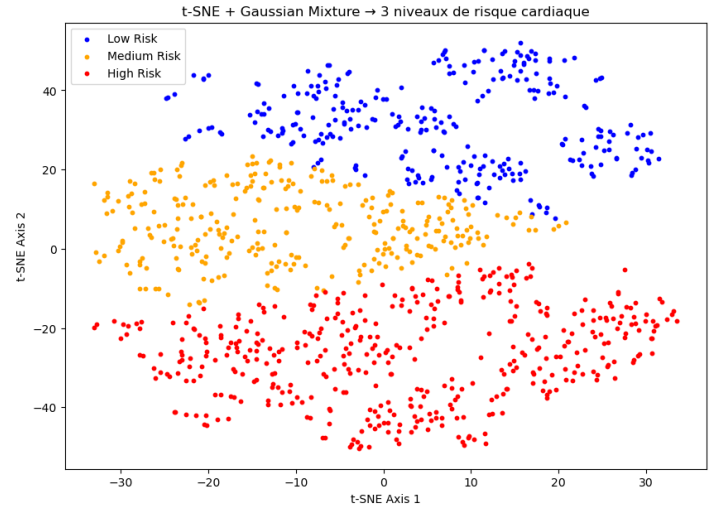


Figure 5: t-SNE + Gaussian Mixture : 3 niveaux de risque cardiaque

The high-risk Smurfs (red cluster) occupy a compact region in the embedding space suggesting that individuals with elevated heart failure risk share common physiological and behavioural characteristics. The low-risk population forms a separate cluster, while the medium-risk group lies between the two, forming a smooth risk gradient.

The mean predicted risk per cluster confirms the separation observed in t-SNE:

Risk Group	Mean Risk Score
Low Risk	0.038
Medium Risk	0.056
High Risk	0.125

Table 11: Average predicted risk within the three identified clusters.

5.2 Feature Profiles of Each Risk Group

Feature	Low Risk	Medium Risk	High Risk
Age	116.34	117.16	118.76
Blood Pressure	104.73	105.70	112.68
Cholesterol	105.26	107.52	119.51
Hemoglobin	14.26	14.27	14.15
Height	7.62	7.58	7.60
Potassium	4.30	4.38	4.27
Vitamin D	29.94	30.39	30.17
Weight	97.58	99.56	114.59

Table 12: Comparison of average physiological measurements across risk groups.

High-risk Smurfs therefore present:

- **Higher blood pressure** (+7.9% compared to medium risk)
- **Higher cholesterol** (+11% compared to medium risk)
- **Substantially higher body weight** (+15–17% compared to other groups)

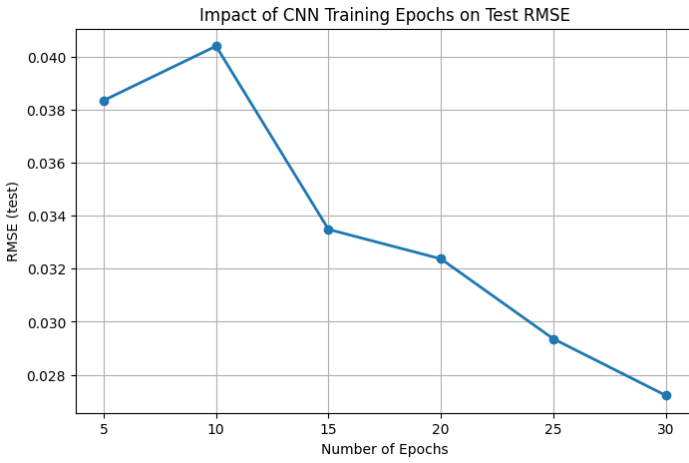


Figure 3: Impact of Training Epochs

Best performance achieved at 30 epochs, suggesting the CNN benefits from prolonged optimization despite the small dataset (N=1000)

4.3 Final Multimodal Model Performance

After integrating CNN-based image embeddings with the selected tabular features, we retrained the best nonlinear model identified in Part 2 (XGBoost). The multimodal model achieved a substantial improvement in predictive accuracy.

$$\text{Train RMSE} = 0.0066, \quad R_{\text{train}}^2 = 0.9933$$

$$\text{Test RMSE} = 0.02951, \quad R_{\text{test}}^2 = 0.87084$$

The multimodal model reduces the test RMSE by **32.2%** and increases test R^2 by **+15.12 percentage points**. This represents a substantial improvement in predictive power, demonstrating that heart scans contain complementary structural information.

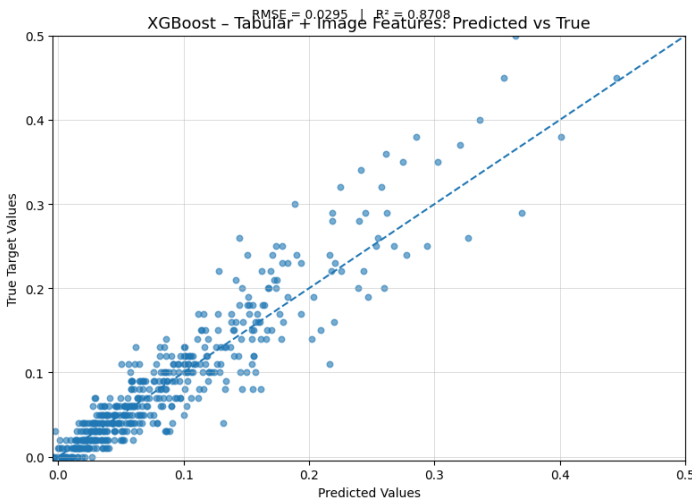


Figure 4: The predicted vs true value

The predicted vs true plot displays an approximately linear diagonal pattern, which means that its predictions closely match the actual values.

- **Lower potassium**, consistent with known clinical markers of cardiac stress

5.3 Distribution of Clinical Variables Across Risk Groups

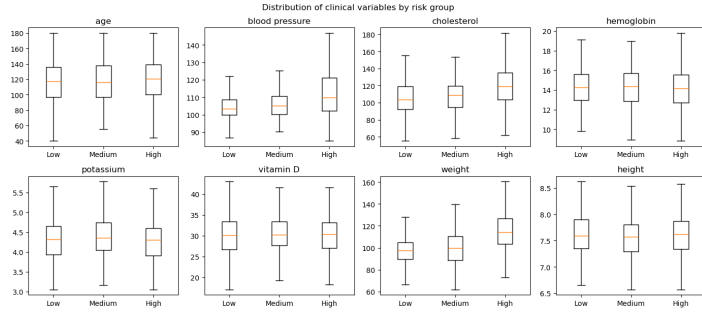


Figure 6: boxplots-risk

The boxplots reveal several clinically meaningful trends:

- **Blood pressure and cholesterol** increase significantly from Low to High risk groups.
- **Weight** is the strongest discriminating variable with high-risk individuals being on average 15–20% heavier.
- **Potassium and hemoglobin** remain relatively stable suggesting weaker predictive influence.
- **Height and age** show minor variation across groups confirming they play a secondary role.

5.4 Interpretation of CNN Heatmap Visualizations

To better understand how the convolutional neural network (CNN) integrates image information in predicting heart failure risk, we generated Grad-CAM-like visual explanations for samples belonging to both low-risk and high-risk groups

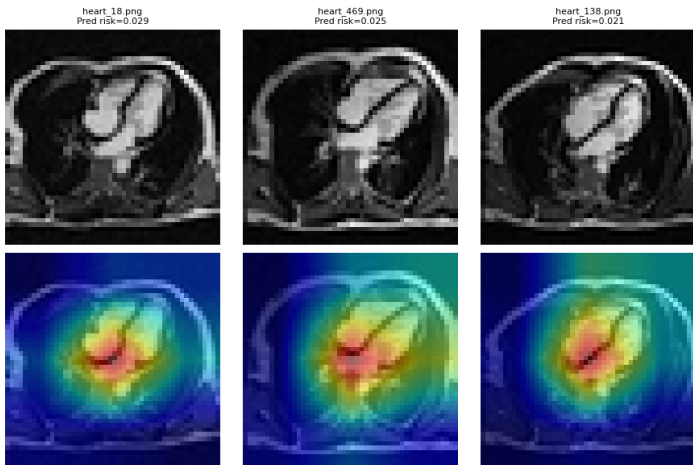


Figure 7: Low-risk examples with corresponding heatmaps

For Smurfs predicted as having a low cardiac risk, the CNN heatmaps show localized attention around well-defined and homogeneous ventricular regions. The highlighted zones correspond to smooth contours, regular myocardial thickness, and globally symmetric heart shapes.

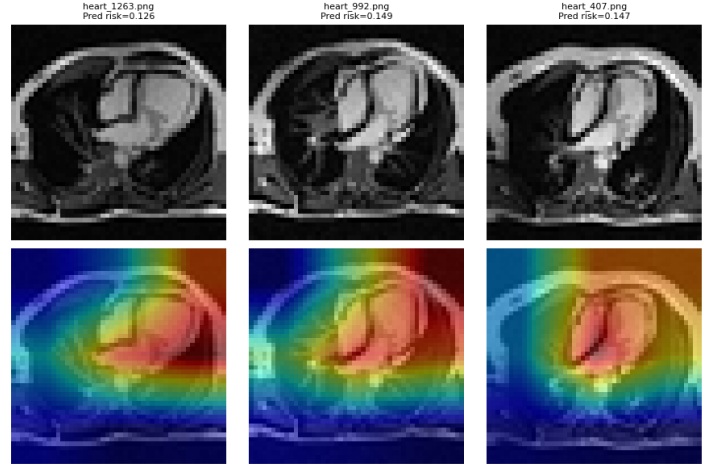


Figure 8: High-risk examples with heatmaps

In contrast, high-risk predictions show substantially different activation patterns. The CNN tends to focus on:

- regions exhibiting shape deformation or asymmetry
- irregular borders or motion artifacts.

6 Final Conclusion

This report presented a comprehensive machine learning approach to predicting heart failure risk in Smurfs, addressing Doctor Smurf's urgent need for an efficient diagnostic tool. Starting from a baseline linear model in Part 1, we progressively enhanced our predictive capabilities through nonlinear models in Part 2, achieving significant improvements with XGBoost. The integration of heart scan image features using CNN in Part 3 led to a substantial performance boost, demonstrating the power of multi-modal learning by combining clinical measurements, lifestyle data, and medical imaging. Part 4 deepened our understanding of heart failure in Smurf society by identifying distinct risk clusters and revealing that high-risk Smurfs are characterized by elevated blood pressure, cholesterol, and weight levels, along with visible heart deformation patterns captured by Grad-CAM visualizations.